

AINE 204 · Term Project

E-commerce Customer Behavior Analysis

Mohamed Emadeldin — AI Tools · Spring 2025/26

Problem Overview

The Challenge

E-commerce businesses struggle to understand **customer spending patterns** and segment their customer base effectively.

- Without data-driven insights, marketing efforts are broad and inefficient
- Businesses miss opportunities to personalize for high-value customers
- Customer churn is difficult to detect without behavioral analysis

Our Solution

Apply AI and ML tools to **predict**, **segment**, and **visualize** customer behavior from a dataset of 351 customers across 6 US cities.

Tool Demonstration — AI Tools Used

Tool	Purpose & Why Chosen
Random Forest	Predict Total Spend; handles mixed data, resistant to overfitting
LabelEncoder	Encode categorical columns; lightweight and pipeline-ready
pandas	Data loading, cleaning, grouping, and transformation
numpy	Numerical computation backbone across the pipeline
matplotlib	All visualizations: bar charts, pie charts, histograms

Tool Demonstration — How They Were Applied

- ➊ **pandas** — Loaded 351-row CSV, filled missing values (mean/mode)
- ➋ **LabelEncoder** — Encoded Gender, City, Membership Type, Discount Applied, Satisfaction Level
- ➌ **Random Forest Regressor** — Trained on 80% data, tested on 20%
- ➍ **numpy** — Supported ratio calculations (Items ÷ Spend) and array operations
- ➎ **matplotlib** — Generated 5 charts: actual vs predicted spend, age pie chart, items/spend histogram, city bar charts, recency bar chart

Key Findings

Analysis	Finding
Spend Prediction	MSE = 85.62 (± 9.25 avg deviation)
Top Age Group	Age 30 — 13.7% of customers
Items per Spend	Mean 0.02, bimodal distribution detected
Highest Spend City	San Francisco — $\approx \$1,460$ avg spend
Most Satisfied City	Chicago & Miami — satisfaction score 2.0
Active Customers	226 (64.4%) purchased in last 30 days
Re-engagement Target	121 Lapsed customers (34.5%)

Standout Finding

San Francisco & New York have the **highest spend** but **lower satisfaction** — a churn risk that needs immediate attention.

Model Comparison — Which AI Model Performs Best?

Model	MSE	RMSE	MAE
Random Forest	85.62	9.25	4.31
Gradient Boosting	99.78	9.99	5.86
Decision Tree	131.83	11.48	4.42
KNN	401.12	20.03	11.65
Linear Regression	890.87	29.85	22.94

Why Random Forest Won:

- Lowest MSE (85.62), RMSE (± 9.25) and MAE (4.31)
- Ensemble of 100 trees reduces overfitting
- Handles mixed data types effectively

Why Linear Regression Failed:

- MSE of 890.87 — over **10x worse** than Random Forest
- Customer spending is **non-linear** — too complex for a straight-line model

Random Forest

MSE: 85.62 — **Best overall**

Gradient Boosting

MSE: 99.78 — Strong alternative

Decision Tree

MSE: 131.83 — Good but inconsistent

Conclusion

Original model choice **validated** by comparison.

Ethical Discussion

Data Privacy

Dataset contains PII (age, city, spend). Requires **anonymization** and GDPR compliance in real-world use.

AI Bias

Model may be biased toward overrepresented cities or age groups, producing skewed predictions for minorities.

Responsible Use

Predictions should **inform** decisions, not automate them. Human oversight must be maintained.

Limitations

- Only 351 records — may not represent all customers
- LabelEncoder implies false ordering between cities
- Static dataset — no real-time behavioral data

Conclusion

Instructor: Dr. Hussain Ahmed

Thank you! Questions?